Department of Electrical, Computer and Biomedical Engineering

ELE 532: Signals and Systems I

Final Examination 14 December 2021 Time: 150 minutes

Instructors: Prof. Javad Alirezaie and Prof. Soosan Beheshti

Answer **all five** questions.
Questions are NOT equally weighted.
If you are in doubt, state your assumptions.
Show all calculations and steps taken to arrive at your answers to get full marks.

First Name: ______________________________

Last Name: ______________________________

Student No.: ______________________________

Section No.: ______________________________

Question	Points	Score
1	15	
2	20	
3	20	
4	25	
5	20	
Total:	100	

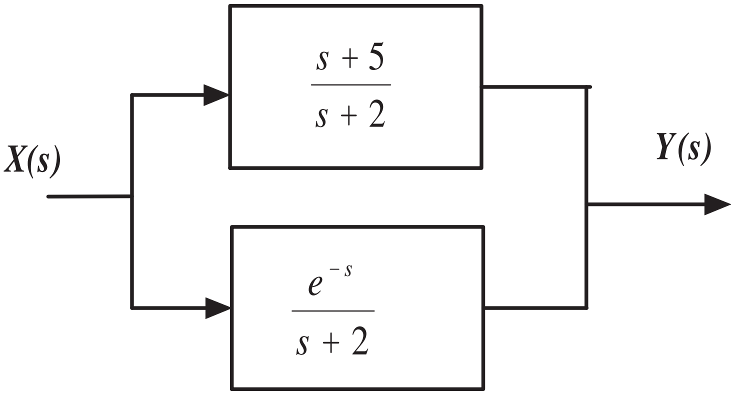
Please also read and sign below:

Consistent with Ryerson's Policy 60: Academic Integrity, I agree that all the assignments, quizzes, project work and exams I complete will represent my work and my work only. I understand that all forms of academic misconduct are prohibited. I also understand that academic misconduct includes, but is not limited to, all forms of cheating, including the use of unauthorized materials, plagiarism, false identification, and forgery. In addition, I understand that it is my duty and my responsibility to familiarize myself with Ryerson's Policy 60: Academic Integrity and inform the instructor if I become aware of any violations to this Honour Code.

Student's signature....................

Question 1 ....

(15 marks total) Two linear systems are connected in parallel as shown in the following figure:



(a) [4 marks] Find the equivalent transfer function $H(s) = \frac{Y(s)}{X(s)}$ of the system and find the region of convergence (ROC) (unilateral case).

(b) [6 marks] Find the impulse response $h(t)$ of the equivalent system.

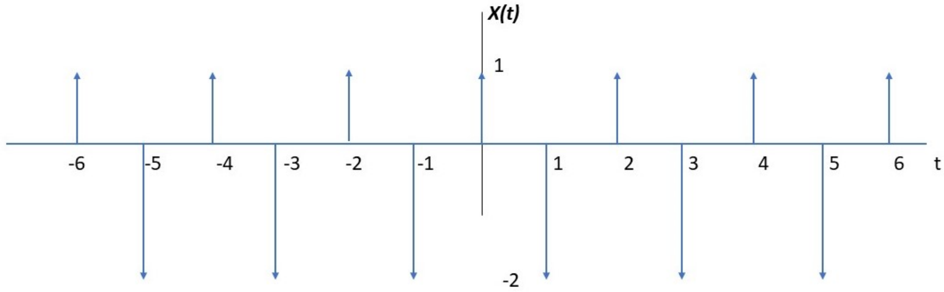
(c) [5 marks] Find the step response of the equivalent system, i.e. find $y(t)$ when $x(t) = u(t)$.

Question 2 . . . .

(20 marks total) For the signal x(t) shown in the below figure,

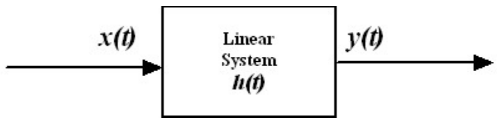
(a) [10 marks] Find the complete exponential Fourier series that converges to x(t) for all time.

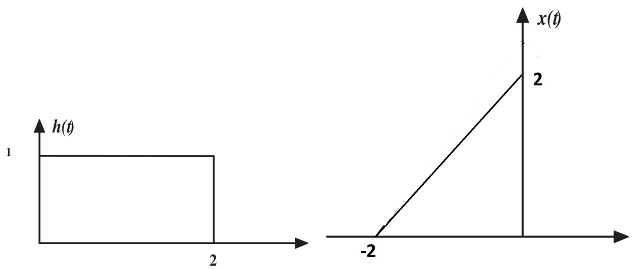
(b) [10 marks] Plot amplitude, $|D_n|$ and phase spectra, $\angle D_n$ of the signal.



Question 3 ....

(20 marks total) A linear system is defined by the following block diagram and the impulse response, $h(t)$, and an input signal $x(t)$ are shown in the following figures:





(a) [12 marks] Use the time domain convolution to find the $y(t)$ the output of the system above if the signal $x(t)$ shown above is applied to the linear system. Plot the signal $y(t)$. (Show all your calculations)

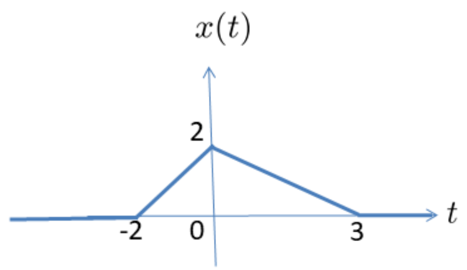
(b) [4 marks] The signal $x(t)$ time reversed to get the signal $x(-t)$. If $x(-t)$ is applied to the same system, find and plot the new zero state response $y_2(t) = h(t) * x(-t)$.

(c) [4 marks] Describe what happens when $h(t)$ is also time reversed that is $y_3(t) = h(-t) * x(-t)$, plot the signal $y_3(t)$.

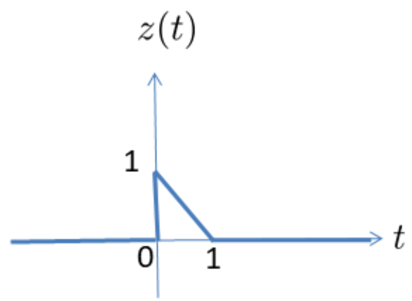
Question 4 . . . .

(25 marks total) The four parts of this question can be answered independently.

Consider the following signal $x(t)$



(a) [7 marks] Write $x(t)$ as a function of provided $z(t)$ (Fig. below). Use that relationship to write $X(j\omega)$ as a function of $Z(j\omega)$.



(b) [7 marks] Draw the first derivative of $x(t)$, $x'(t)$. Find the fourier transform of this first derivative. Use this fourier transform to provide the fourier transform of $x(t)$.

(c) [6 marks] Draw the second derivative of $x(t)$, $x''(t)$. Find the fourier transform of this second derivative. Use this fourier transform to provide the fourier transform of $x(t)$.

(d) [5 marks] Show the prove for the following relationship: If $x(t)$ has fourier transform $Xj\omega)$ then delayed version of $x(t)$, $y(t) = x(t - t_0)$ has fourier transform $Y(j\omega) = e^{-j\omega t_0}X(j\omega)$.

Question 5 . . . .

(20 marks total)

(a) [3 marks] Draw an impulse train $\Delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$ and show the product of this impulse train with an arbitrary $x(t)$.

(b) [2 marks] Draw an example of $X(j\omega)$ that is bandlimited by the value of B (i.e. has only nonzero values between $-B$ and B). For simplicity your $X(j\omega)$ can be real with no phase.

(c) [5 marks] Show the fourier transform of the product from part (a), $y(t) = x(t)\Delta_T(t)$, for $B = 10\pi$ and $T = 0.01$.

(d) [10 marks] Suggest a method that we recover $x(t)$ from the sampled $y(t)$ in part (c). Also for what other values of T this recovery is possible? explain.